ANALYTICAL CUMULATIVE EXAM
December 3, 1994

Trace Metal Determination via Electrochemical Stripping Analysis

Below you will answer questions regarding Electrochemical Stripping Analysis of Metals from aqueous solutions. Use the data in the attached tables and information to help in answering the questions. Here are some key points that should also help you:

- Mercury can be used as an electrode in electrochemical experiments. Some forms include a hanging drop at the end of a glass capillary containing the mercury. The drop is submerged into an aqueous solution and at the opposite end of the capillary one makes electrical contact to the mercury. Another form of mercury electrode is a mercury film that is deposited or wiped onto another electrode substance and then submerged into the aqueous solution of interest.

- Mercury forms amalgams with many metals, such as Cu, In, Pb, Cd, Tl, Zn, etc.

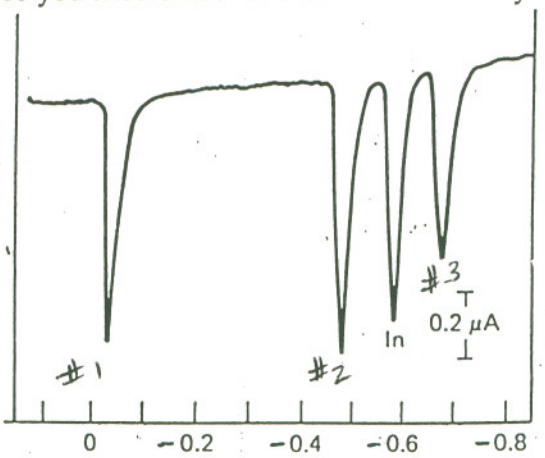
(15 pts) 1. (a) Define *mercury amalgam.*
(b) What is the standard reduction potential (E°) of copper amalgamation
- vs. the Normal Hydrogen Electrode *and*
- vs. the Standard Calomel Electrode?

(30 pts) 2. Steps to carry out a stripping analysis....
(a) Suggest an electrochemical way, using a potentiostat and a mercury film electrode or hanging mercury drop electrode, to obtain a mercury amalgam of Cu^o from an electrolyte solution containing Cu^{2+} ions. Use the redox potential in your answer to #1b in your description. Also include the applied potential function and the reference electrode that you would use.
(b) Describe how you could potentiostatically remove (or "strip") the Cu from the amalgam (given the redox potential in #1b). Include in your description the applied potential function and the reference electrode that you would use.
(c) How could you *quantitatively* relate the current signal obtained during the stripping step to the concentration of Cu^{2+} in the solution?

(10 pts) 2. (a) What is the relationship between the time of the amalgam formation and the detection limits of the Cu^{2+} determination? Why?
(b) What *else* could be done to improve detection limits, besides changing time of amalgamation? Why?

(45 pts) 3. A group of graduate students run stripping voltammetry experiments on a "standard" solution of 0.1 M KNO_3, 2 x 10^{-7} M In^{3+}, and 2 x 10^{-7} M Cu^{2+}. The analysis shows more peaks than just for In and Cu.
(Show all work and calculations to the following questions.)

(15 pts) (a) Identify the peaks #1,#2, and #3 in the following stripping voltammetry curve of a metal film electrode in the "standard" solution;
(*Note:*
- the potential is not scanned positive enough to see the oxidation of Hg
- the peak potentials in the figure below do not quite line up with the redox potentials in the tables, even after adjusting for the different reference electrode. However, if you can figure out how one of the peak potentials is shifted with respect to the redox potential in the table, then all you have to do is apply the same shift to the other potentials in the table.
- there are only a very limited number of amalgamated metal redox potentials in the attached table, so you should not have too much difficulty identifying peaks.)



E/V *vs.* SCE

(30 pts) (b) What is (are) the major source(s) of contamination, given the following information?

Here is what the students wrote in their lab notebook:
Reference electrode = SCE.
Prepared Hg film electrodes by depositing Hg from a solution of 1 mM $Hg(NO_3)_2$ dissolved in DI water.
Prepared 0.1 M KNO_3, diluted to mark with tap water.
Made up a 500 mL solution of 0.1 M KNO_3 containing both 2 x 10^{-7} M $In(NO_3)_3$ and 2 x 10^{-7} M Cu^{2+}. It was prepared by weighing out $In(NO_3)_3$, dissolving Cu metal in 1 ml conc. HNO_3, and then diluted to mark with 0.1 M KNO_3 electrolyte.

The tap water is known to contain:
20 ppb Pb, 0.1 ppb Zn

All reagents were obtained from company X, with the following analysis information:
conc. HNO_3: 20 ppb Pb, 1 ppb In
$Hg(NO_3)_2$: 1 ppm Pb
KNO_3: 1 ppm Cd
$In(NO_3)_3$: 1 ppm Cd
Cu metal: 1 ppm Cd, 1 ppm Zn

Data that could be useful

RT/F = 0.02569 V; F/RT = 38.92 V^{-1}; F = 96485 Coul/mol e^-

An NHE electrode referenced against an NHE electrode is 0.000 V.
An SCE electrode referenced against an NHE electrode is -0.2412 V.

Reaction	E^o vs. NHE
$Cd^{2+} + 2e^- \rightarrow Cd$	-0.4026
$Cd^{2+} + 2e^- \rightarrow Cd(Hg)$	-0.3521
$Cu^{2+} + 2e^- \rightarrow Cu$	+0.3402
$Cu^{2+} + 2e^- \rightarrow Cu(Hg)$	+0.345
$Pb^{2+} + 2e^- \rightarrow Pb$	-0.1263
$Pb^{2+} + 2e^- \rightarrow Pb(Hg)$	-0.1205
$Zn^{2+} + 2e^- \rightarrow Zn$	-0.7628
$Zn^{2+} + 2e^- \rightarrow Zn(Hg)$	-0.690

International Atomic Weights

Element	Symbol	Atomic number	Atomic weight	Element	Symbol	Atomic number	Atomic weight
Actinium	Ac	89	(227)	Mercury	Hg	80	200.59
Aluminum	Al	13	26.9815	Molybdenum	Mo	42	95.94
Americium	Am	95	(243)	Neodymium	Nd	60	144.24
Antimony	Sb	51	121.75	Neon	Ne	10	20.183
Argon	Ar	18	39.948	Neptunium	Np	93	(237)
Arsenic	As	33	74.9216	Nickel	Ni	28	58.70
Astatine	At	85	(210)	Niobium	Nb	41	92.906
Barium	Ba	56	137.34	Nitrogen	N	7	14.0067
Berkelium	Bk	97	(247)	Nobelium	No	102	(254)
Beryllium	Be	4	9.0122	Osmium	Os	76	190.2
Bismuth	Bi	83	208.980	Oxygen	O	8	15.9994
Boron	B	5	10.811	Palladium	Pd	46	106.4
Bromine	Br	35	79.909	Phosphorus	P	15	30.9738
Cadmium	Cd	48	112.40	Platinum	Pt	78	195.09
Calcium	Ca	20	40.08	Plutonium	Pu	94	(244)
Californium	Cf	98	(249)	Polonium	Po	84	(210)
Carbon	C	6	12.01115	Potassium	K	19	39.102
Cerium	Ce	58	140.12	Praseodymium	Pr	59	140.907
Cesium	Cs	55	132.905	Promethium	Pm	61	(145)
Chlorine	Cl	17	35.453	Protactinium	Pa	91	(231)
Chromium	Cr	24	51.996	Radium	Ra	88	(226)
Cobalt	Co	27	58.9332	Radon	Rn	86	(222)
Copper	Cu	29	63.54	Rhenium	Re	75	186.2
Curium	Cm	96	(245)	Rhodium	Rh	45	102.905
Dysprosium	Dy	66	162.50	Rubidium	Rb	37	85.47
Einsteinium	Es	99	(254)	Ruthenium	Ru	44	101.07
Erbium	Er	68	167.26	Samarium	Sm	62	150.35
Europium	Eu	63	151.96	Scandium	Sc	21	44.956
Fermium	Fm	100	(252)	Selenium	Se	34	78.96
Fluorine	F	9	18.9984	Silicon	Si	14	28.086
Francium	Fr	87	(223)	Silver	Ag	47	107.870
Gadolinium	Gd	64	157.25	Sodium	Na	11	22.9898
Gallium	Ga	31	69.72	Strontium	Sr	38	87.62
Germanium	Ge	32	72.59	Sulfur	S	16	32.064
Gold	Au	79	196.967	Tantalum	Ta	73	180.948
Hafnium	Hf	72	178.49	Technetium	Tc	43	(99)
Helium	He	2	4.0026	Tellurium	Te	52	127.60
Holmium	Ho	67	164.930	Terbium	Tb	65	158.924
Hydrogen	H	1	1.00797	Thallium	Tl	81	204.37
Indium	In	49	114.82	Thorium	Th	90	232.038
Iodine	I	53	126.9044	Thulium	Tm	69	168.934
Iridium	Ir	77	192.2	Tin	Sn	50	118.69
Iron	Fe	26	55.847	Titanium	Ti	22	47.90
Krypton	Kr	36	83.80	Tungsten	W	74	183.85
Lanthanum	La	57	138.91	Uranium	U	92	238.03
Lawrencium	Lw	103	(257)	Vanadium	V	23	50.942
Lead	Pb	82	207.19	Xenon	Xe	54	131.30
Lithium	Li	3	6.942	Ytterbium	Yb	70	173.04
Lutetium	Lu	71	174.97	Yttrium	Y	39	88.905
Magnesium	Mg	12	24.312	Zinc	Zn	30	65.37
Manganese	Mn	25	54.9380	Zirconium	Zr	40	91.22
Mendelevium	Mv	101	(256)				

Numbers in parentheses indicate mass number of most stable known isotope.